

# Essential Summary

## A Generalized Harmonic Function Perturbation Method for Determining Limit Cycles and Homoclinic Orbits of Helmholtz–Duffing Oscillator

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*Journal of Sound and Vibration*, Vol. 332, No. 21, 5508–5522 (2013) DOI: 10.1016/j.jsv.2013.05.007

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## 1 Background and Motivation

The Helmholtz–Duffing oscillator, governed by  $\ddot{x} + c_1\dot{x} + c_2x^2 + c_3x^3 = 0$ , appears in ship dynamics, oscillations of the human eardrum, one-dimensional structural systems with initial curvature, and heavy symmetric gyroscopes. The asymmetric quadratic term  $c_2x^2$  breaks the symmetry present in classical Duffing oscillators, introducing qualitatively different dynamical behavior.

When perturbed by nonlinear damping, the oscillator

$$\ddot{x} + c_1\dot{x} + c_2x^2 + c_3x^3 = \varepsilon f(\mu, x, \dot{x}) \quad (1)$$

exhibits limit cycles and homoclinic bifurcations. Unlike the Duffing oscillator (where  $c_2 = 0$ ), the Helmholtz–Duffing oscillator requires a fundamentally different analytical treatment because its potential energy is asymmetric, and its exact solution cannot be expressed through standard elliptic functions alone.

This paper introduces the **generalized harmonic function perturbation (GHFP) method**—a novel perturbation framework that, for the first time, provides a unified treatment for both limit cycle determination and homoclinic orbit prediction in the damped Helmholtz–Duffing oscillator.

## 2 Core Innovation: Quadratic Generalized Harmonic Function

### 2.1 Nonlinear Time Transformation

The key innovation is the introduction of a nonlinear time transformation:

$$\frac{d\varphi}{dt} = \Phi(\varphi), \quad \Phi(\varphi + \pi) = \Phi(\varphi) \quad (2)$$

that converts the time-domain equation into an angular-domain form over the symmetric interval  $[0, \pi]$ .

### 2.2 Solution Construction

The periodic solution is constructed as a **quadratic generalized harmonic function**:

$$x(t) = a \cos^2 \varphi(t) + b \quad (3)$$

where  $a, b \in \mathbb{R}$ . This representation differs fundamentally from prior approaches that used  $x = a \cos \varphi + b$ . The  $\cos^2 \varphi$  form automatically captures the asymmetry induced by the quadratic term  $c_2x^2$ .

For homoclinic orbits, the solution takes the alternative form:

$$x(t) = a \sin^2 \varphi(t) + b \quad (4)$$

### 2.3 Determining $a$ and $b$

Substituting Eq. (3) into the undamped oscillator and using Eq. (2), the angular velocity  $\Phi(\varphi)$  is obtained as:

$$\Phi(\varphi) = \sqrt{\frac{2[V(a+b) - V(a \cos^2 \varphi + b)]}{(x'(\varphi))^2}} \quad (5)$$

where  $V(x) = \int_0^x g(u)du$  and  $g(x) = c_1x + c_2x^2 + c_3x^3$ .

The constants  $a$  and  $b$  are determined by:

$$\int_{a+b}^b g(u)du = 0 \quad (6)$$

and  $a+b = x(0)$  (the initial condition). For homoclinic orbits, the saddle point  $h$  satisfies  $g(h) = 0$ , and the initial condition  $c$  is determined by  $\int_h^c g(x)dx = 0$ .

## 3 The GHFP Perturbation Procedure

### 3.1 Perturbation Expansion

When  $\varepsilon \neq 0$ , the solution parameters are expanded in powers of  $\varepsilon$ :

$$a = a_0 + \varepsilon a_1 + \cdots, \quad \Phi = \Phi_0 + \varepsilon \Phi_1 + \cdots \quad (7)$$

Substituting into Eq. (1) and equating coefficients of like powers of  $\varepsilon$  yields a hierarchy of equations:

$$\varepsilon^0 : \quad \frac{d}{d\varphi}(\Phi_0^2 x'_0) + g(x_0) = 0 \quad (8)$$

$$\varepsilon^1 : \quad \frac{d}{d\varphi}(\Phi_0 \Phi_1 x'_0 + \Phi_0^2 x'_1) + g'(x_0)x_1 = f(\mu, x_0, \Phi_0 x'_0) \quad (9)$$

### 3.2 Key Assumption

The critical assumption that simplifies the perturbation procedure: **the  $n$ th-order approximate solution  $x_n$  has the same functional form as the generating solution:**

$$x_n = a_n \cos^2 \varphi \quad (\text{or } x_n = a_n \sin^2 \varphi \text{ for homoclinic orbits}) \quad (10)$$

This eliminates the cumbersome integration required in classical perturbation methods.

### 3.3 Central Result: Bifurcation Parameter Prediction

The first-order consistency condition (obtained by integrating the  $\varepsilon^1$  equation over  $[0, \pi]$ ) yields:

$$\mu = - \frac{\int_0^\pi \Phi_0 f_0(\mu, x_0, \Phi_0 x'_0) x'_0 d\varphi}{\int_0^\pi \Phi_0 \frac{\partial f_0}{\partial \mu} x'_0 d\varphi} \quad (11)$$

This formula directly gives the critical value of the control parameter  $\mu_c$  at which homoclinic bifurcation occurs—previously obtainable only through numerical shooting methods.

## 4 Key Results

### 4.1 Example 1: $c_1 = 1, c_2 = 0.5, c_3 = 1$

Consider the damped Helmholtz–Duffing oscillator:

$$\ddot{x} + x + 0.5x^2 + x^3 = \varepsilon(\mu + \mu_1 x + \mu_2 x^2)\dot{x} \quad (12)$$

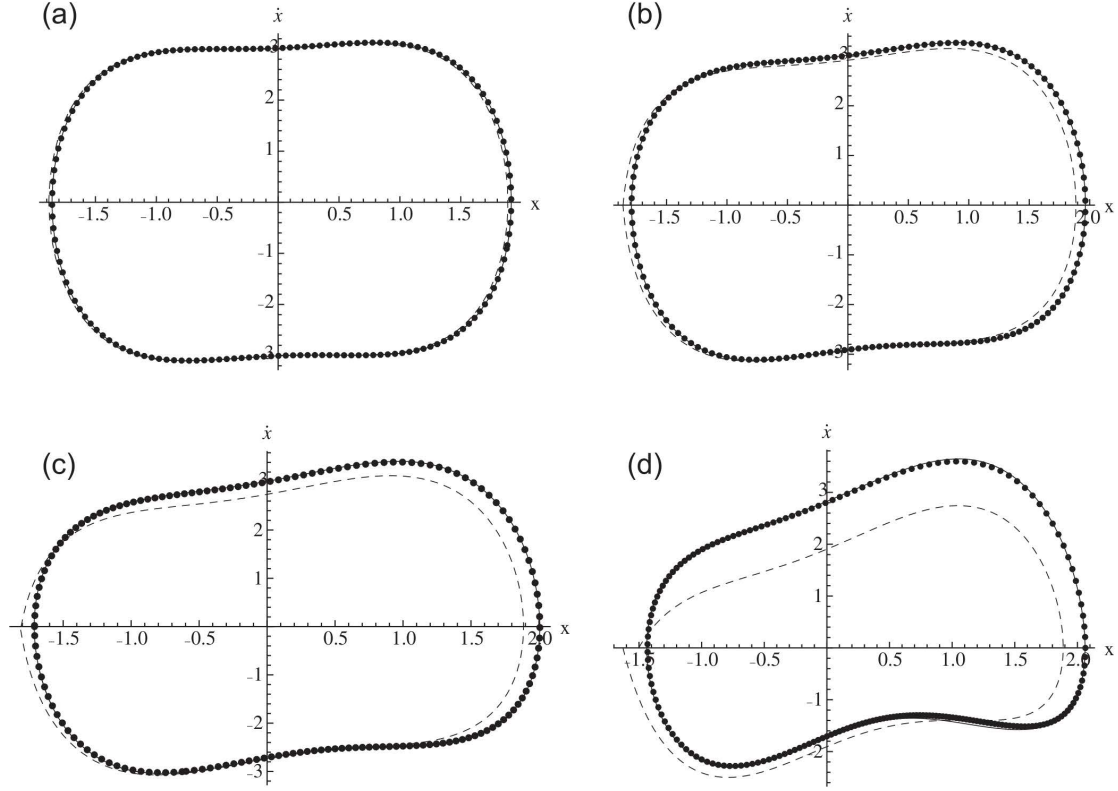


Figure 1: **Figure 1.** (Original Fig. 1) Limit cycles of oscillator (12) for  $\varepsilon = 0.1, 0.3, 0.5, 1$ . Solid lines: Runge–Kutta method. Dotted lines: GHFP method. Excellent agreement at all perturbation levels demonstrates the method’s accuracy.

#### 4.2 Example 2: $c_1 = 1$ , $c_2 = -1$ , $c_3 = 2$ —Homoclinic Orbits

$$\ddot{x} + x - x^2 + 2x^3 = \varepsilon(\mu + \mu_2 x^2)\dot{x} \quad (13)$$

This oscillator possesses a saddle point at  $x = 0$  and two additional equilibrium points, enabling homoclinic bifurcation. The critical parameter values predicted by the GHFP method are in close agreement with numerical results.

## 5 Significance and Impact

1. **Foundation of the GHFP family:** This paper establishes the quadratic generalized harmonic function framework—the foundation upon which all subsequent GHFP and MGHP methods are built. The insight that  $x = a \cos^2 \varphi + b$  (rather than  $a \cos \varphi + b$ ) is the correct basis function for asymmetric oscillators proved essential.
2. **First analytical homoclinic bifurcation prediction:** Before this work, predicting the critical parameter for homoclinic bifurcation in strongly nonlinear oscillators required numerical shooting. The GHFP method provides the first analytical formula.
3. **Unified treatment of limit cycles and global bifurcations:** The same framework handles both local (limit cycle) and global (homoclinic/heteroclinic) dynamical phenomena, eliminating the need for separate analytical approaches.

## 6 Limitations

- The method is limited to first-order perturbation accuracy; higher-order extensions would improve precision for larger  $\varepsilon$ .

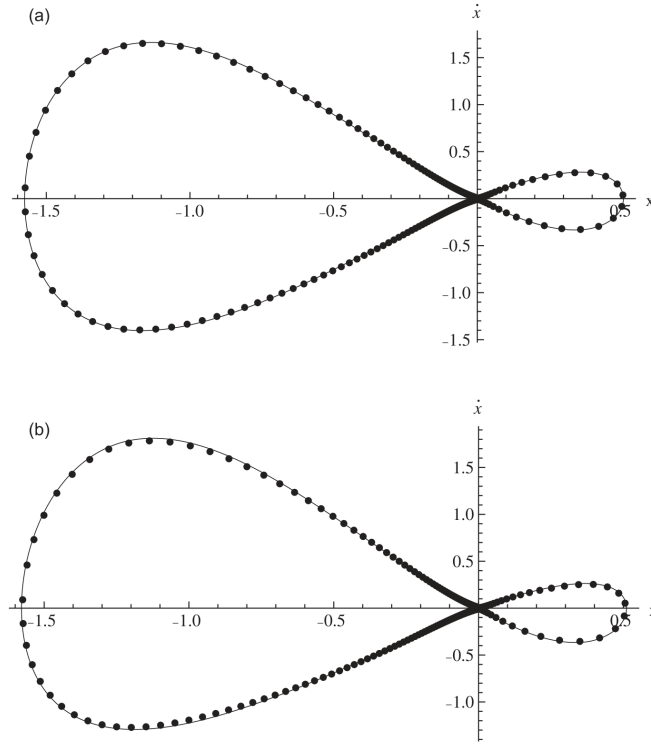


Figure 2: **Figure 2.** (Original Fig. 5) Homoclinic orbits of oscillator (13) for  $\varepsilon = 1, 2$ . Solid lines: Runge–Kutta. Dotted lines: GHFP method. The method accurately captures homoclinic orbit geometry even at  $\varepsilon = 2$ .

- The approach assumes the nonlinear damping is small ( $\varepsilon \ll 1$ ); strongly damped systems require alternative treatments.
- Only single-degree-of-freedom oscillators are considered; multi-degree-of-freedom extensions remain an open challenge.

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